MVA-PRISM Track 2

Biomarker-gated drug repurposing for candidate BUB1B-mediated Mosaic Variegated Aneuploidy

Lead proposal: pharmacologic NAD+ augmentation with approved nicotinic acid (niacin) as a testable strategy to increase residual BUBR1 abundance through the NAD+-SIRT2-BUBR1 axis. Secondary, biomarker-gated proposal: rapalog-class mTORC1 inhibition only if patient-derived cells show pathway hyperactivation.

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Competition: Rare Disease, Real Kid: The MVA Hackathon 2026 - Track 2

Research-use statement. The medicines discussed here are repurposing hypotheses for experimental validation. This report does not recommend treatment of the child, does not establish safety or efficacy in MVA, and does not propose off-label administration outside an appropriately approved research or clinical protocol.

1. Executive Summary

Track 1 prioritized a candidate biallelic BUB1B model consisting of GRCh38 chr15:40209701 T>G (NM_001211.6:c.2210T>G; p.Leu737Ter) and chr15:40220612 T>G (NM_001211.6:c.3006T>G; p.Asn1002Lys). Both are heterozygous PASS calls in the challenge VCF; phase is unresolved. The first is a ClinVar-pathogenic nonsense variant for MVA1. The second is a rare missense variant in the BUBR1 kinase domain and remains a variant of uncertain significance. Therefore, the drug proposal is explicitly conditional on confirmation that the missense allele produces residual, unstable or functionally impaired BUBR1 protein.

The central therapeutic hypothesis is not to suppress cell division globally, but to increase the amount or functional persistence of residual BUBR1 protein from the missense allele. Human BUBR1 missense mutations near the kinase domain can undergo accelerated proteasomal turnover, while BUBR1 restoration rescues checkpoint function in MVA patient cells. Separately, the NAD+-dependent deacetylase SIRT2 has been shown to regulate BUBR1 abundance; NAD+ precursor administration increased BUBR1 abundance in vivo in mice. These observations create a pharmacologically testable bridge to nicotinic acid (niacin), an approved medication and NAD+ precursor.

A second, mechanistically distinct branch addresses downstream signaling. A published MVA mouse model carrying a BUBR1 truncating allele near the patient candidate truncation showed hyperactive mTORC1 signaling and sarcopenia. Because the proband has low muscle mass/skeletal muscle atrophy, mTORC1 should be measured in patient-derived cells. If phospho-S6/p70S6K is elevated, a rapalog can be evaluated ex vivo as a pathway-normalization strategy. Everolimus is favored for translational benchmarking because of established pediatric regulatory experience, while sirolimus provides the closest mechanistic comparator to rapamycin literature.

Key innovation: this is a biomarker-gated rescue framework. A drug advances only if it reverses a pre-specified molecular defect in patient-derived cells: BUBR1 abundance, spindle checkpoint performance, chromosome missegregation, or a validated downstream signaling abnormality.

2. Case-Linked Molecular Mechanism

BUB1B encodes BUBR1, a core spindle-assembly-checkpoint and kinetochore regulatory protein. Biallelic BUB1B dysfunction causes MVA1, a recessive chromosome-instability syndrome characterized by whole-chromosome missegregation, mosaic aneuploidy, growth restriction, congenital abnormalities and childhood cancer predisposition. The challenge phenotype includes rhabdomyosarcoma, nephrocalcinosis, short stature, failure to thrive, skeletal muscle atrophy, prematurity, severe low birth weight/IUGR and recurrent parental miscarriages.

The candidate p.Leu737Ter allele is predicted loss-of-function. The p.Asn1002Lys allele lies inside the canonical human BUBR1 protein-kinase domain (UniProt annotates residues 766-1050). A particularly relevant precedent is the published MVA combination of a BUBR1 truncating allele (X753) and a kinase-domain missense allele (L1012P). The positions are not identical to this case, and functional equivalence must not be assumed, but the topology is unusually similar: a truncation near residue 737/753 paired with a late kinase-domain missense near residue 1002/1012.

Mechanistically, MVA-associated BUBR1 variants can reduce total BUBR1 abundance by two routes: loss of transcript from truncating alleles and increased turnover/misfolding of kinase-domain missense proteins. Low BUBR1 compromises the spindle checkpoint and stable kinetochore-microtubule attachment, allowing premature anaphase, lagging chromosomes, micronuclei and whole-chromosome aneuploidy.

Level	Predicted defect in this case	Measurable readout
Variant	Truncating allele + kinase-domain missense allele	Orthogonal confirmation; parental or long-read phasing
Protein	Reduced/unstable BUBR1 abundance	Western blot / quantitative immunoassay

Level	Predicted defect in this case	Measurable readout
Checkpoint	Impaired SAC and error correction	Mitotic arrest response; kinetochore BUBR1; lagging chromosomes
Genome	Chromosome missegregation / mosaic aneuploidy	Micronucleus assay, metaphase counts, interphase FISH or single-cell DNA
Downstream	Possible allele-specific mTORC1 hyperactivation	p-S6, p-p70S6K, p-4EBP1

3. Candidate 1 - Nicotinic Acid (Niacin): NAD+-SIRT2-BUBR1 Rescue

Repurposing rationale. Nicotinic acid is an approved medication and a biochemical precursor for NAD+. In human cells, nicotinic acid can increase intracellular NAD+ through nicotinate phosphoribosyltransferase (NAPRT). SIRT2 is an NAD+-dependent deacetylase. In a BubR1 aging model, increasing NAD+ with nicotinamide mononucleotide increased BUBR1 abundance in vivo, and SIRT2 overexpression increased BUBR1 and improved lifespan. The proposal is to test whether an approved NAD+ precursor can reproduce the BUBR1-abundance component of this biology in patient-derived MVA cells.

Why this may fit the candidate genotype. A truncating allele cannot be pharmacologically restored by niacin. The opportunity lies in the second allele: if p.Asn1002Lys produces residual BUBR1 that is unstable or rapidly turned over, even a modest increase in steady-state protein could improve checkpoint reserve. This is directly falsifiable. If the missense allele produces no protein, or if increasing NAD+ does not increase BUBR1 abundance or chromosome-segregation fidelity, the hypothesis fails.

Important mechanistic caveat. BUBR1 acetylation is site-specific and complex. K668 deacetylation by SIRT2 has been associated with increased BUBR1 abundance, whereas K250 acetylation participates in mitotic checkpoint timing and protection from APC/C-dependent degradation. Therefore, simply increasing NAD+ cannot be assumed to be beneficial. This is precisely why an ex vivo rescue experiment is required before any translational consideration.

Element	Pre-specified test	Go criterion
Target engagement	NAD+ metabolome and SIRT2-dependent BUBR1 regulation	Biochemical evidence of NAD+ augmentation without loss of BUBR1
Protein rescue	Total BUBR1 and kinetochore-localized BUBR1	Reproducible increase versus vehicle in patient cells
Functional rescue	SAC response, lagging chromosomes, micronuclei	Improved mitotic fidelity without excessive cytotoxicity
Genomic rescue	Aneuploidy burden over serial passages	Directionally reduced chromosome missegregation / mosaic aneuploidy

Clinical status and safety. Niacin is FDA-recognized as a prescription lipid-modifying medicine. That approval does not establish efficacy or pediatric safety for MVA. Known risks include hepatotoxicity, flushing, glucose dysregulation and hyperuricemia; these issues are particularly relevant when considering a medically complex child. No dose is proposed in this report.

4. Candidate 2 - Biomarker-Gated mTORC1 Inhibition

A published BUBR1 MVA model found that a truncating X753 allele produced hyperactive mTORC1 signaling in skeletal muscle and was associated with progeroid/sarcopenic phenotypes. The present candidate truncation p.Leu737Ter is

nearby, and the proband has low muscle mass/skeletal muscle atrophy. This creates a testable - but not assumed - downstream mechanism.

The correct translational sequence is: first measure p-S6, p-p70S6K and p-4EBP1 in patient-derived cells; only if mTORC1 is reproducibly hyperactive should a rapalog be tested. Everolimus is a practical translational candidate because it is an approved mTOR inhibitor with pediatric regulatory experience in tuberous sclerosis complex; sirolimus is a useful mechanistic comparator because rapamycin/sirolimus is the canonical compound used in mTOR biology.

This branch is deliberately secondary. mTOR inhibition may worsen growth, suppress immunity and impair wound healing, and it has not been shown to correct the primary BUBR1 spindle-checkpoint defect in MVA. A decrease in p-S6 without improvement in mitotic fidelity would be interpreted as pathway modulation, not disease correction.

5. Proposed Validation Pipeline

Stage	Experiment	Decision
A. Genotype	Confirm both BUB1B variants and establish phase by parental segregation or validated long-read/linked phasing.	If variants are in cis, reopen causal search before drug work.
B. Baseline biology	Patient fibroblasts (or another ethically obtained primary cell type): quantify BUBR1 abundance, SAC function, missegregation, micronuclei and aneuploidy.	Require a reproducible cellular defect relative to controls.
C. Niacin test	Evaluate target engagement and BUBR1 abundance, then chromosome-segregation readouts. Use an NAD-raising reagent used in prior BUBR1 studies as a mechanistic positive control where appropriate.	Advance only with concordant molecular + functional rescue.
D. mTOR gate	Measure p-S6/p70S6K/4EBP1. If elevated, test rapalog pathway normalization and mitotic endpoints.	No biomarker elevation = no rapalog branch.
E. Replication	Repeat in independent cultures/passages; blinded image scoring; pre-specified effect direction.	Require reproducibility before translational planning.
F. Translation	Independent ethics/regulatory review, pediatric pharmacology, oncology/genetics input and prospective protocol.	Human administration only after appropriate approvals.

6. Scientific Rigor, Innovation, Impact and Scalability

Judging criterion	How this proposal addresses it
Scientific rigor (35%)	Variant-to-protein-to-pathway chain; explicit uncertainty about phase and VUS status; pre-specified rescue readouts; falsification criteria; no clinical leap from computational prediction.
Potential impact (25%)	If BUBR1 abundance can be pharmacologically increased, the concept targets a core molecular deficit rather than a single symptom. A biomarker-gated strategy may also improve clinical stratification.
Innovation (25%)	Repurposes an approved NAD+ precursor through a specific SIRT2-BUBR1 abundance mechanism and pairs it with allele-aware downstream signaling analysis.
Scalability (15%)	The workflow can be repeated across BUB1B genotypes: measure residual BUBR1, test abundance rescue, quantify chromosome segregation, then layer allele-specific pathway biomarkers.

7. Investigator and Translational Execution Capacity

Dr. Bary G. Bigay Mercedes is a physician with postgraduate training spanning clinical genomics, rare diseases, molecular biology, reproductive medicine, genomic oncology, bioinformatics and omics analysis. His curriculum vitae documents experience in high-throughput genome sequencing, statistical genetics and clinical NGS interpretation, and identifies him as Executive Medical Director of Instituto ChromoMED. It also describes institutional leadership of clinical genetics, genomic medicine protocols and translational research programs, as well as postgraduate teaching in genomic bioinformatics.

This background is relevant to the proposal because Track 2 requires more than a drug name: it requires a pathway hypothesis that can be operationalized. Instituto ChromoMED can serve as a translational coordination site for confirmatory genotyping, patient phenotyping, biomarker collection, cell-based validation and research protocol development, subject to the scope of its validated laboratory capabilities and external collaborators.

Regulatory boundary: the existence of clinical and research infrastructure does not by itself authorize experimental treatment. Any administration of a repurposed medicine for MVA would require independent ethics/IRB review, applicable national regulatory authorization, informed consent/assent, pediatric safety oversight, appropriate oncology/genetics/pharmacology collaboration and a prospectively defined monitoring plan.

8. Limitations

The major uncertainty remains the second BUB1B allele. p.Asn1002Lys is not functionally validated, and the two candidate variants are not yet proven to be in trans. The NAD+-BUBR1 evidence is derived from mechanistic cell and mouse studies, not MVA clinical trials. Nicotinic acid has not been shown to increase BUBR1 in MVA cells. Likewise, mTORC1 hyperactivation is allele-specific in published models and cannot be inferred solely from the presence of a nearby truncating allele. These limitations are not reasons to abandon the hypotheses; they define the experiments required to test them.

The proposal therefore prioritizes a **rapid ex vivo falsification strategy** rather than empirical patient exposure. A negative rescue result is scientifically informative and should terminate that branch.

9. Conclusion

MVA-PRISM Track 2 proposes a precision repurposing strategy anchored in the candidate BUB1B genotype. The lead hypothesis is that approved nicotinic acid can augment NAD+ sufficiently to enhance the SIRT2-BUBR1 axis and increase residual BUBR1 from the missense allele, thereby improving spindle-checkpoint reserve and chromosome-segregation fidelity. The secondary hypothesis is that a rapalog may normalize an allele-specific mTORC1 abnormality if - and only if - that abnormality is demonstrated in patient-derived cells.

The strength of the proposal is its decision architecture: genotype confirmation -> baseline molecular defect -> drug target engagement -> functional chromosome rescue -> independent replication -> regulated translation. This makes the approach testable in the present case and scalable to other individuals with BUB1B-mediated MVA.

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12. MVA Hackathon 2026 controlled-access challenge materials: clinical phenotype document and WGS VCF for proband WGS_EX2312012. Used under challenge data-use rules.

AI/data-handling disclosure

OpenAI ChatGPT Team workspace, GPT-5.6 Sol, was used for literature synthesis, analytical cross-checking and preparation of this submission. Challenge genomic data were handled under the competition data-use framework. The report contains participant-generated hypotheses and selected variant coordinates, not redistributed raw genomic or clinical source files.